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CAUSAL RELATIONSHIP BETWEEN EXPORTS, IMPORTS AND ECONOMIC GROWTH: TIME SERIES STUDY FOR REPUBLIC OF UZBEKISTAN

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Abstract. *This study investigates the causal nexus of imports, exports, and economic growth in Uzbekistan using time series evidence and advanced econometric techniques. The empirical analysis begins with unit root testing by employing the Augmented Dickey-Fuller (ADF) test, followed by the Johansen cointegration test to establish long-run equilibrium relationships between the variables. A Vector Error Correction Model (VECM) is employed to test short- and long-run causality. The results reveal unidirectional causality from imports to exports, changes in import activity significantly influence export performance. The study further finds bidirectional (bicausal) relations between imports and economic growth and between exports and economic growth. The implications of these findings are that both trade components do play a significant role in the economic development of Uzbekistan. The results emphasize the importance of balanced trade policy that benefits both growth in imports and promotion of exports in order to achieve long-term economic growth.*

Keywords: *imports, exports, economic growth, Augmented Dickey-Fuller (ADF) test, a Vector Error Correction model (VECM).*

O‘ZBEKISTON RESPUBLIKASI UCHUN VAQT QATORLARI ASOSIDA EKSPORT, IMPORT VA IQTISODIY O‘SISH O‘RTASIDAGI BOG‘LIQLIKNI MIDQORIY TADQIQ QILISH

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Annotatsiya. *Ushbu tadqiqot O‘zbekiston Respublikasida import, eksport va iqtisodiy o‘sish o‘rtasidagi sabab-oqibat bog‘liqligini vaqt qatori ma’lumotlari va zamonaviy ekonometrik uslublar asosida o‘rganadi. Empirik tahlil Augmented Dickey-Fuller (ADF) testi orqali o‘zgaruvchilarning birlik ildizga ega ekanligi yoki yo‘qligini aniqlashdan boshlandi, so‘ngra o‘zgaruvchilar o‘rtasidagi uzoq muddatli muvozanat aloqalarini aniqlash uchun Johansen kointegratsiya testi qo‘llanildi. Qisqa va uzoq muddatli sababiy bog‘liqliklarni aniqlash uchun Vektor Xatolik Tuzatish Modeli (VECM) ishlatildi. Natijalar importdan eksportga qarab bir yo‘nalishli sababiy bog‘liqlik mavjudligini ko‘rsatadi, ya’ni import faoliyatidagi o‘zgarishlar eksport natijalariga sezilarli ta’sir ko‘rsatadi. Tadqiqot shuningdek, import va iqtisodiy o‘sish hamda eksport va iqtisodiy o‘sish o‘rtasida ikki yo‘nalishli (o‘zaro) bog‘liqlik mavjudligini aniqlaydi. Ushbu natijalarning ahamiyati shundaki, savdoning har ikkala tarkibiy qismi-import* *va eksport-O‘zbekistonning iqtisodiy rivojlanishida muhim rol o‘ynaydi. Natijalar uzoq muddatli iqtisodiy o‘sishga erishish uchun import o‘sishini ham, eksportni rag‘batlantirishni ham qamrab oluvchi muvozanatli savdo siyosatining muhimligini ta’kidlaydi.*

Kalit so‘zlar: *import, eksport, iqtisodiy o‘sish, kengaytirilgan Dikki-Fuller testi (ADF), xatolikni tuzatish vektor modeli (VECM).*

ПРИЧИННО-СЛЕДСТВЕННАЯ СВЯЗЬ МЕЖДУ ЭКСПОРТОМ, ИМПОРТОМ И ЭКОНОМИЧЕСКИМ РОСТОМ: ИССЛЕДОВАНИЕ НА ОСНОВЕ ВРЕМЕННЫХ РЯДОВ ДЛЯ РЕСПУБЛИКИ УЗБЕКИСТАН

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Аннотация. *Данное исследование рассматривает причинно-следственную взаимосвязь между импортом, экспортом и экономическим ростом в Узбекистане с использованием данных временных рядов и современных эконометрических методов. Эмпирический анализ начинается с проверки на единичный корень с помощью расширенного теста Дики-Фуллера (ADF), за которым следует тест на коинтеграцию Йохансена для выявления долгосрочных равновесных отношений между переменными. Для анализа краткосрочной и долгосрочной причинной связи используется модель коррекции ошибок векторного типа (VECM). Результаты показывают однонаправленную причинную связь от импорта к экспорту, то есть изменения в импортной активности существенно влияют на показатели экспорта. Также установлены двунаправленные (обоюдные) связи между импортом и экономическим ростом, а также между экспортом и экономическим ростом. Полученные выводы свидетельствуют о том, что обе составляющие внешней торговли играют значительную роль в экономическом развитии Узбекистана. Результаты подчеркивают важность сбалансированной торговой политики, направленной как на рост импорта, так и на стимулирование экспорта, с целью обеспечения устойчивого экономического роста в долгосрочной перспективе.*

Ключевые слова: *импорт, экспорт, экономический рост, тест дополненного Дики-Фуллера (ADF), модель векторной коррекции ошибок (VECM).*

Introduction.

Trade has long been regarded as one of the prime sources of economic growth owing to the integration of national economies into the global economic system. Through trade, a nation enhances the efficient allocation of resources in an open market, enabling the home economy to utilize its comparative advantages more effectively and benefit from scale economies. Moreover, trade facilitates the diffusion of technological innovations and enhances competitiveness both domestically and internationally. All these cumulatively bring about increased productivity and economic optimization (Bernard et al., 2003; Bernard & Jensen, 2004; Haq et al. 2021; Haq & Zhu, 2019; Ahmad et al. 2024). Exports, in particular, are an important source of foreign exchange earnings, which ease pressures on a country's balance of payments. Growth in exports can also trigger employment generation and economic diversification. The process of export-led growth can also enhance technological capabilities to cater to domestic as well as foreign market demands. Empirical studies by Kemal et al. (2002) found a positive link between exports and economic growth in certain South Asian countries. Similarly, Zestos and Tao (2002), in a study, found the presence of bidirectional causality between exports and GDP and between imports and GDP in the Canadian context. A number of studies have confirmed the hypothesis that growth in exports has a significant contribution to economic growth in general (Ullah et al., 2009; Andrew, 2015; Saaed and Hussain, 2015). The causality of this relationship is, nevertheless, contentious. There is also evidence of a bidirectional or feedback relationship between the two variables, as required by Ramos (2001) and Liu et al. (2002). Despite the strong theoretical and empirical support of the trade-growth link, not all studies arrive at the same conclusion. For instance, Asafu-Adjaye and Chakraborty (1999) found no long-run causal relationship between exports and economic growth in India. Likewise, Yuhong et al. (2010) concluded that export activity was not an important determinant of economic growth. The same findings were reached by Aicha (2015) for the Moroccan case, as there was no proof of causality in any direction. The intricate relationship between the export and import activities of a country and how they reflect on economic performance is a central area of interest for economists, policymakers, and academic researchers. This is in great measure because economic growth is, worldwide, one of the most integrated indices of a nation's development and general welfare (Han, & Haq, 2017; Khan & Khan, 2021). In the past two decades, the globe has experienced dramatic globalization, increased trade integration, and a shift in global trade patterns (Sokolov-Mladenović et al., 2016). One of the aspects that have dominated economic development discourse is the Export-Led Growth (ELG) hypothesis, which states that the growth of exports is a key force behind economic growth (Jordaan & Eita, 2007). Historically, the focus of much of the literature has been on the ways in which export stimulates economic performance. The ELG theory is promoted by advocates who feel that exports trigger growth by virtue of countries—particularly small or developing economies—being able to exploit economies of scale, ease foreign exchange constraints necessary for bringing in capital and intermediate goods, raise aggregate productivity due to increased competition in world markets, and stimulate technological advances through learning-by-doing processes (Mahadevan & Suardi, 2008).

But the importance of imports has also been increasingly seen, especially in the context of endogenous growth models, where imports supply the most important channels for technology and knowledge spillovers from developed economies to developing economies (Ramos, 2001). Foreign technological importation, commonly embedded in intermediate goods and services like machinery and equipment, can significantly increase labor productivity and facilitate the adoption and adaptation of new ways of production by domestic industries. Workers, through exposure to these sophisticated tools, learn new techniques and assist in the unbundling and assimilation of embodied technology. Baharumshah and Rashid (1999) emphasized the crucial developmental role of such technology-burdened imports. Here, Awokuse (2007) cautions that emphasizing exports alone as the sole determinant of growth is a partial account. Leaving out the contribution of imports to the process of growth risks overlooking their critical contribution to furnishing innovation, stimulus to productivity, and general economic transformation. There are numerous empirical studies supporting the concept of two-way causal relationship between imports and economic growth (Kogid et al., 2011; Andrew, 2015; Aicha, 2015), which presents a positive feedback effect. For instance, the study conducted by Saaed and Hussain (2015) in Tunisia is more specific. The outcome indicated unidirectional causality from imports to economic growth, from economic growth to exports, and from imports to exports. They concluded that while exports directly influence economic performance, imports indirectly influence growth by providing indirect access to important capital goods, ideas, and technological inputs. This means that a good growth strategy should recognize the complementary roles played by both imports and exports in driving sustainable economic growth. Italicized. This study explores the interdependent dynamic and causal relationships between exports, imports, and economic growth in Uzbekistan as a bid to cast further insight into how international trade influences the economic performance of the nation. In light of continuing efforts in Uzbekistan to develop towards a more open and market-oriented economy, understanding dynamics of trade components to

GDP is essential for policy decision-making. Drawing on time series econometric techniques like stationarity tests, cointegration tests, and the Vector Error Correction Model (VECM), this research investigates short-run and long-run causal relationships between these critical macroeconomic variables. The findings will be useful to policymakers who would like to catalyze enduring economic growth via intelligent trade policy reforms.

Research methodology.

Time series analysis comprises statistical methods for analyzing data points that have been collected or recorded at discrete time intervals. It finds especial use in finance, economics, and environmental science. Important elements of this kind of analysis are stationarity tests, cointegration, and causal relationships between variables. One of the simplest processes in time series analysis is testing for stationarity, which refers to a time series whose characteristics are not dependent on the moment the series is being sampled. The Augmented Dickey-Fuller (ADF) test can be routinely applied to determine if a time series contains a unit root and is therefore non-stationary (Dickey & Fuller, 1979). The null hypothesis for the ADF test is that the series is a unit root, and the alternative hypothesis is that the series is stationary. If the null hypothesis is rejected, then the time series is deemed stationary and no further differencing is required.

When dealing with two or more non-stationary series, researchers must determine whether there is a long-run equilibrium relationship between them. It is here that the cointegration principle becomes relevant. Two or more time series are said to be cointegrated if a linear combination of them is stationary, even though the individual series themselves are not stationary (Engle & Granger, 1987). The Johansen cointegration test is employed for this in multivariate applications in most situations. It tests for the number of cointegrating relations and has a more general treatment of dynamics than the Engle-Granger two-step procedure.

If there is cointegration, then the appropriate model framework would be the Vector Error Correction Model (VECM). In contrast to a Vector Autoregression (VAR) model for stationary series, a VECM can deal with non-stationary but cointegrated series. The VECM not only specifies the short-run dynamics in terms of differenced terms, but also encompasses long-run equilibrium relations in terms of an error correction term. This model framework permits causality tests, short run (using lagged differences) and long run (using the error correction term). Granger causality tests can be performed in the VECM framework to establish whether or not a variable is useful for forecasting another (Granger, 1981). This study analyzed the time series data on exports, imports and gross domestic product from 1995 to 2023 and is collected from World Bank (2025).

Results.

The output of Augmented Dickey-Fuller (ADF) test shows that the three variables—log of GDP, exports, and imports—are non-stationary at level because their test statistics (−1.957, −2.218, and −1.784) are greater than the 5% critical value. But when first differenced, the test statistics (−4.832, −5.024, and −4.712) are all significant at 1% level and therefore the variables are stationary. That means the variables are integrated of order one, or I(1). They become stationary only after first-differencing, as is necessary for carrying out cointegration and Vector Error Correction Model (VECM) tests. ADF test Results are displayed in Table 1.

Johansen cointegration test results, based on the Max-Eigenvalue statistic, reject the null hypotheses for both $r = 0$ and $r = 1$ as the test statistics (28.925 and 14.992) are greater than the 5% critical values (21.131 and 14.264, respectively). The test statistic (3.212) at $r = 2$ is, however, less than the critical value (3.841), leading to the failure to reject the null hypothesis. This shows the presence of a single cointegrating vector among the variables. Exports, imports, and economic growth in Uzbekistan therefore have a long-run equilibrium relationship, confirming they do move together over time. Cointegration Test results for trace statistics and maximum eigen statistics are given in Table 2 and Table 3 respectively.

Table 1:

Augmented Dickey-Fuller (ADF) Test Results

Variable	Level (t-Statistic)	First Difference (t-Statistic)	Stationarity at Level	Stationarity at First Difference	Conclusion
Ln(GDP)	-1.957	-4.832***	No	Yes	I(1)
Ln(Exports)	-2.218	-5.024***	No	Yes	I(1)
Ln(Imports)	-1.784	-4.712***	No	Yes	I(1)

Note: Critical values at 1% = -3.610, 5% = -2.939, 10% = -2.607.
*** denotes significance at the 1% level.

Table 2:

Johansen Cointegration Test Results (Trace Test)

Null Hypothesis	Eigenvalue	Trace Statistic	5% Critical Value	Conclusion
r = 0	0.652	47.129	29.797	Reject H_0
r ≤ 1	0.423	18.204	15.495	Reject H_0
r ≤ 2	0.129	3.212	3.841	Reject H_0

Table 3:

Johansen Cointegration Test Results (Max-Eigenvalue Test)

Null Hypothesis	Max-Eigen Statistic	5% Critical Value	Conclusion
r = 0	28.925	21.131	Reject H_0
r = 1	14.992	14.264	Reject H_0
r = 2	3.212	3.841	Reject H_0

The VECM estimates confirm strong short-run and long-run causal relationships between economic growth, imports, and exports in Uzbekistan. In the equation for exports as the dependent variable, the Wald test indicates statistically significant short-run causality from imports (χ^2 = 9.763, p = 0.002), and the error correction term (ECT = -0.342, t = -3.912***) confirms a strong long-run adjustment mechanism. This suggests that imports have a significant influence over driving exports in the short run and long run, perhaps through imported inputs to enable export-oriented production. For imports, causality is tested in the short run from GDP (χ^2 = 5.298, p = 0.021), and the evidence of the large ECT (t = -2.846**) proves long-run causality from GDP to imports.

Table 4:

Causality Results

Dependent Variable	Short-Run Causality (Wald χ^2, p-value)	Significant Short-Run Causality From	Error Correction Term (ECT)	ECT t-Statistic	Long-Run Causality	Conclusion
ΔlnExports	χ^2 = 9.763, p = 0.002	Imports	-0.342	-3.912***	Yes	**Imports → Exports** (short and long run)
ΔlnImports	χ^2 = 5.298, p = 0.021	GDP	-0.187	-2.846**	Yes	**Bicausality** with GDP
ΔlnGDP	χ^2 = 7.542, p = 0.006	Exports, Imports	-0.298	-4.214***	Yes	**Exports ↔ GDP**, **Imports ↔ GDP**

This suggests that economic growth is equal to increased demand for imports in the long run, and there is also short-run feedback. Finally, in the GDP model, exports and imports are found to be short-run causality (χ^2 = 7.542, p = 0.006), and the ECT (t = -4.214***) confirms long-run causality. Tests confirm strong bidirectional (bicausal) relationships between trade variables and GDP. Overall, the results confirm interdependence of growth and trade in Uzbekistan with a prominent role for imports in export growth and growth.

Conclusion.

The causal relationships among exports, imports, and economic growth were examined for Uzbekistan using time series econometric techniques including the Augmented Dickey-Fuller (ADF) test, Johansen cointegration test, and the Vector Error Correction Model (VECM). The results provided noteworthy information regarding the dynamic relationship between trade and economic performance in the context of a changing and more open economy. The findings reveal a one-way causality from import to export, which further suggests that the flow of imported goods, most significantly capital goods, raw materials, and intermediate inputs, plays a major role in enhancing the country's export ability. Additionally, two-way (bicausal) relationships were identified between economic growth and export, and between economic growth and import. This implies that both import and export not only result from but also are stimulated by the level of economic activity, justifying their central role in Uzbekistan's development trajectory.

These findings have significant policy implications. First, considering the significance of imports in driving export expansion, policymakers must see to it that trade policy does not impede the effective importation of necessary production inputs. Lowering tariff levels on intermediate and capital goods, simplifying customs rules, and enhancing logistical infrastructure could increase the productivity and international competitiveness of Uzbekistan's export industries. Second, the causality between trade and economic growth highlights the necessity of one integrated trade-growth strategy. Trade promotion policies must be aligned with economic development goals overall. This encompasses the encouragement of sectors of high export potential through investment promotion, skill creation, and promotion of innovation. Similarly, growth promotion policies such as infrastructure building, industry modernization, and ease of finance improvement can also increase trade performance. Third, export and import diversification is needed. Uzbekistan must reduce its dependence upon a narrow list of commodities by expanding value-added sectors and penetrating new markets. This can improve the country's capacity to withstand foreign shocks and shifts in the international market. Finally, strengthening regional and global trade partnerships can further advance the trade and development prospects of Uzbekistan. Active participation in trade agreements and regional integration processes can provide greater market access, induce foreign capital inflow, and facilitate technology transfer. Ultimately, trade plays a central and multifaceted role in the economic growth of Uzbekistan. A well-crafted and effective trade policy—imbued in structural reforms and institutional backing—can realize the maximum potential of exports and imports to drive long-term and inclusive growth.

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